Stochastic Models and Simulation

Week 3: is the simulated staffing effect larger than simulation noise?

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2026/27



One run gives one possible morning

Last week, one parametric morning gave:

$$\text{5.78 min with 16 agents} \quad \longrightarrow \quad \text{4.15 min with 17 agents},$$

a reduction of 1.63 minutes in mean waiting time.

The same model with new random outcomes gives:

Repetition	16 agents	17 agents	Reduction
1	2.23	1.44	0.79
2	1.44	0.66	0.78
3	0.66	0.32	0.34
4	0.75	0.32	0.43
5	6.54	3.94	2.60

Question: what is the expected reduction, and how accurately have we estimated it?

Three different sources of uncertainty

Operational variability: mornings differ in demand, arrival times and service times. *Describe this spread as part of the result.*

Model and input uncertainty: arrival, service, routing or caller-exit assumptions may be wrong. *Running the same model more often cannot fix this.*

Today's focus — Monte Carlo error: with a finite number of repetitions, the estimated model mean changes when the experiment is repeated with new random draws. *Repeating independent mornings reduces this error.*

This week quantifies Monte Carlo error while keeping the Week 2 model fixed.

Define the quantity before estimating it

For simulated morning r and staffing level c, let

$$Y_c^{(r)} = \text{mean wait among callers arriving from 09:00 to 11:00.}$$

Compare the two staffing levels within that morning:

$$D_r = Y_{16}^{(r)} - Y_{17}^{(r)}.$$

A positive D_r is a reduction in waiting from adding the seventeenth agent.

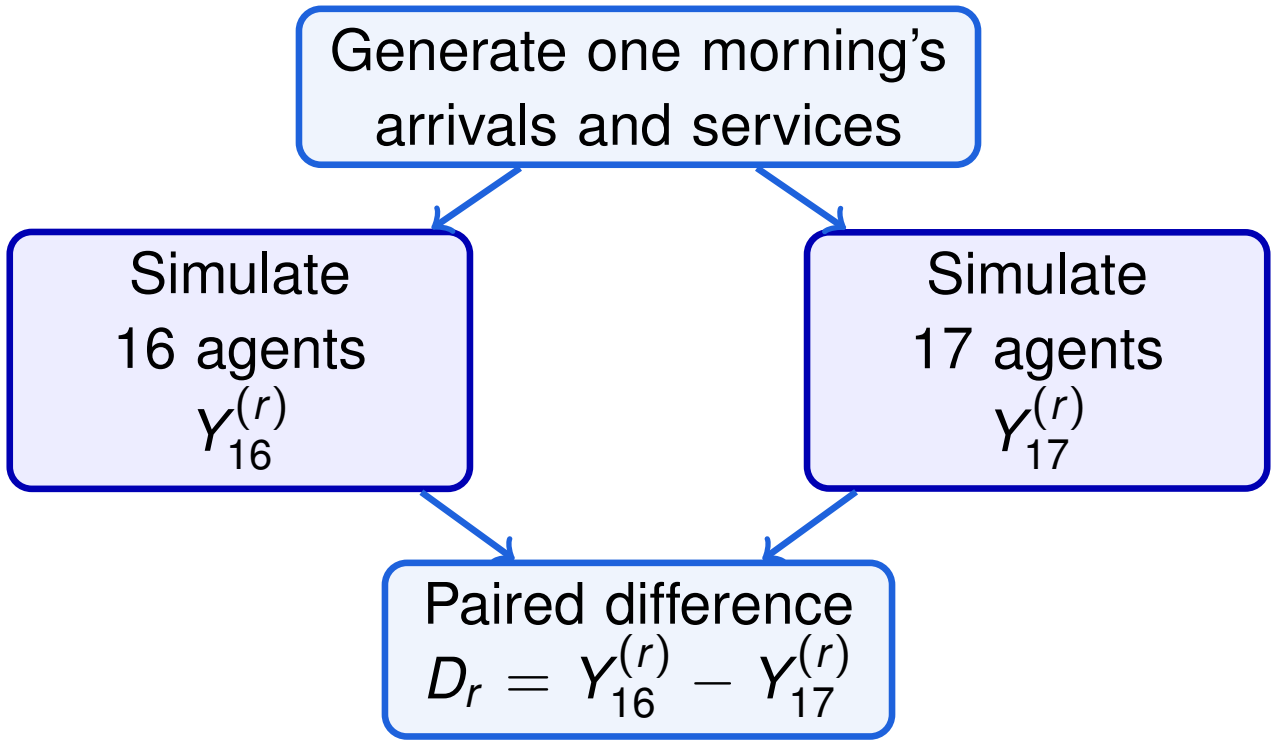
Our target is

$$\delta = \mathbb{E}[D_r],$$

the expected reduction across mornings generated by the stated model. After R independent simulated mornings, we estimate it with

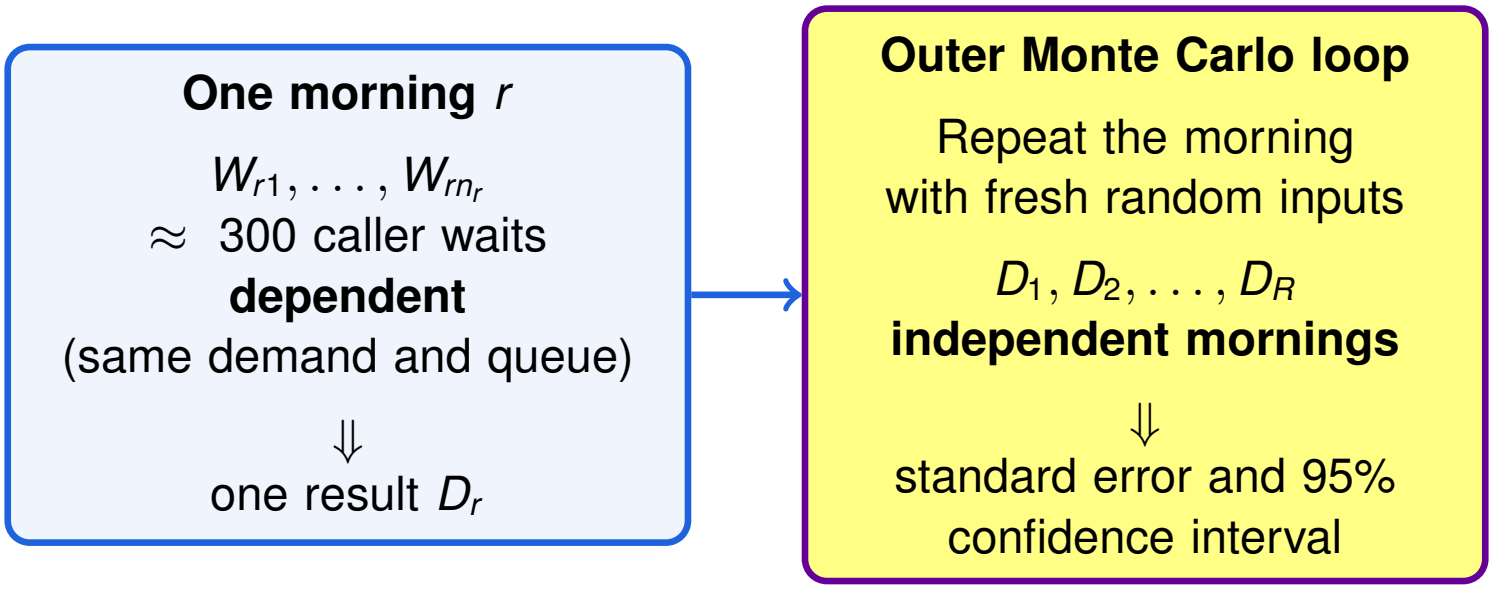
$$\bar{D} = \frac{1}{R} \sum_{r=1}^{R} D_r.$$

One repetition produces one paired difference



The two simulations use **identical arrivals and service requirements**. Only the number of agents changes.

The outer loop creates independent repetitions



Stationary alternative: one much longer run can estimate a long-run mean, but warm-up and dependence then need separate treatment. Here we estimate performance over a finite 09:00–11:00 morning.

What 200 repetitions produced

Simulate 200 independent mornings; paired 16/17 comparison:

	16 agents	17 agents	Reduction D_r
Mean wait (min)	4.35	2.75	1.60
Standard deviation (min)	4.79	3.52	1.38
10th percentile (min)			0.12
90th percentile (min)			3.80

The 1.38-minute standard deviation and the 0.12–3.80 percentile range describe how D_r varies between mornings. They do not measure uncertainty in the estimated mean $\bar{D}$.

Seeds: arrival stream `20260301`; service stream `20260302`. Each morning starts empty at 09:00, as in Week 2.

Standard deviation and standard error answer different questions

Let s_D be the sample standard deviation of $D_1, \ldots, D_R$.

Standard deviation s_D

$$s_D = 1.38 \text{ min}$$

Describes how the staffing benefit varies from one simulated morning to another.
More repetitions estimate this spread better; they do not make mornings less variable.

Standard error of the mean

$$\mathrm{SE}(\bar{D}) = \frac{s_D}{\sqrt{R}} = 0.098 \text{ min}.$$

Describes the Monte Carlo precision of the estimated expected benefit $\bar{D}$.
It decreases as the number of independent mornings grows.

The law of large numbers and the central limit theorem

The target and its estimate after R independent mornings are

$$\delta = \mathbb{E}[D_r], \qquad \bar{D}_R = \frac{1}{R}\sum_{r=1}^{R} D_r.$$

Law of large numbers

$$\bar{D}_R \longrightarrow \delta \qquad \text{as } R \to \infty.$$

It explains why the estimated mean settles toward the expected reduction.

Central limit theorem

$$\frac{\bar{D}_R - \delta}{s_D/\sqrt{R}} \approx \mathcal{N}(0,1).$$

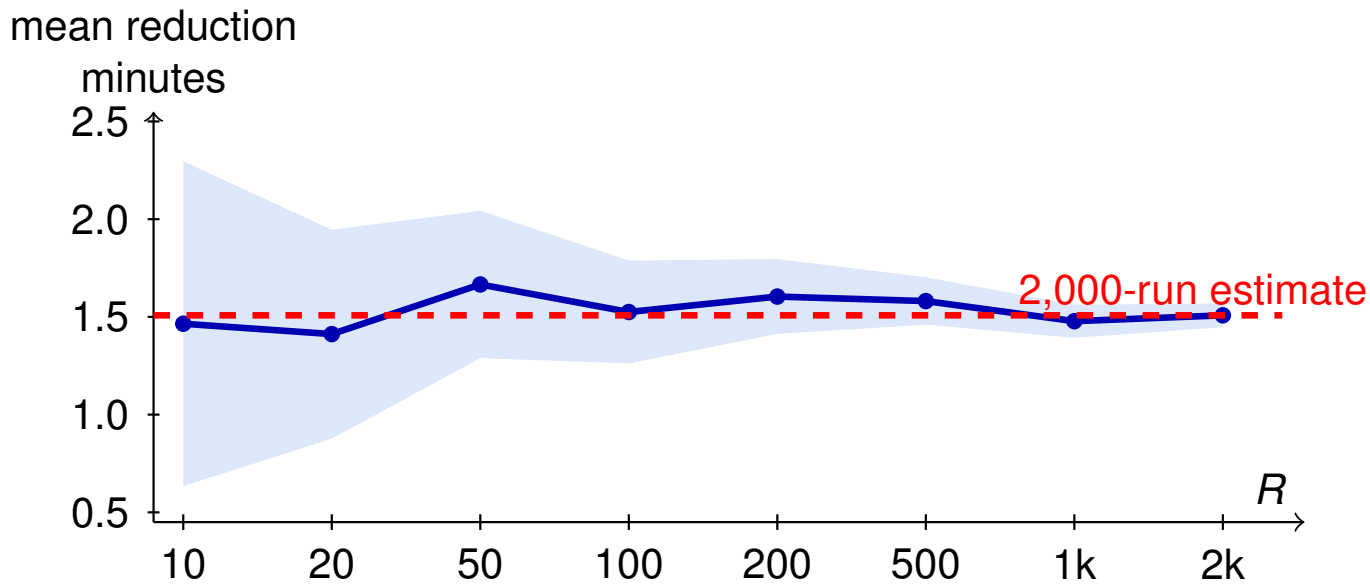
It gives the approximate 95% interval

$$\bar{D}_R \pm 1.96\frac{s_D}{\sqrt{R}}.$$

With $R = 200$, the interval is $[1.41, 1.80]$ minutes. It lies above zero.

What the two limit results look like in the experiment

Target: $\delta = \mathbb{E}[D_r]$.



LLN: the blue running mean settles toward δ. **CLT:** the shaded approximate 95% interval narrows at rate $1/\sqrt{R}$. The dashed red line is the 2,000-run estimate, used only as a visual reference for the unknown δ.

Plan repetitions from a pilot run

If a pilot gives standard deviation s_D, target a confidence-interval half-width h with

$$R \approx \left(\frac{1.96 s_D}{h} \right)^2 .$$

The first 30 repetitions gave $s_D = 1.33$ minutes:

Target half-width h	Planned repetitions
0.50 min	28
0.25 min	110
0.20 min	171
0.10 min	683

Halving the target error requires roughly four times as many repetitions. Recheck the achieved half-width after the planned run because the pilot standard deviation is itself estimated.

Pairing removes irrelevant noise

In the 200-repetition parametric experiment:

$$\operatorname{corr}\left(Y_{16}^{(r)}, Y_{17}^{(r)}\right) \approx 0.99.$$

Busy mornings are difficult under both staffing levels. Pairing uses that positive correlation.

Analysis of the staffing difference	Standard error	95% half-width
Paired, using D_r	0.098 min	0.192 min
If treated as independent samples	0.420 min	0.824 min

The simulation effort is unchanged. The experimental design makes the comparison much more precise.

Repeat the analysis with the empirical generator

Each row uses 200 independent paired repetitions.

Arrival generator	Wait 16	Wait 17	Mean reduction and 95% CI
Parametric model	4.35	2.75	1.60 [1.41, 1.80]
bootstrap	5.50	3.65	1.86 [1.68, 2.03]

Both generators support the same qualitative result for the stated scenario. Their effect sizes differ.

Narrow Monte Carlo intervals do not resolve the difference between models. That difference is a sensitivity to the arrival generator and must be reported separately.

What the evidence supports after Week 3

Supported under the stated model

- estimate the expected 16-to-17-agent effect;
- quantify its Monte Carlo error;
- plan enough independent repetitions;
- compare two arrival generators separately.

Still required for a bank decision

- service-time evidence;
- parameter uncertainty;
- routing, staffing schedules and abandonment;
- an operational benefit threshold and staffing cost.

Use an LLM to audit the experiment design

Give the LLM the replication-loop pseudocode, not private records or student work:

Audit whether this simulation experiment estimates a paired 16-versus-17-agent effect. Check whether repetitions use fresh random draws, whether both staffing levels receive identical arrivals and services, whether the replication unit is the morning, and whether the reported standard error uses the paired differences. For each concern, propose a minimal reproducible check and its expected outcome.

Your work:

1. trace each claim to a specific line or stored output;
2. calculate the expected result of every proposed check independently;
3. run useful checks and retain them;
4. record any unsupported claim or correction.

A confident audit is not evidence until its checks have known, independently verified outcomes.

Notebook route

1. Turn the Week 2 experiment into a function that returns one morning-level result.
2. Generate independent mornings and pair the 16- and 17-agent simulations.
3. Inspect the distribution of the paired differences D_r.
4. Separate standard deviation across mornings from standard error of the mean.
5. Build and interpret an approximate 95% confidence interval.
6. Use a pilot run to plan the number of repetitions.
7. Repeat the analysis with the whole-morning bootstrap generator.
8. State the supported conclusion and the remaining model uncertainty.

What we take forward

- One simulation run is one possible system history, not an estimate with known precision.
- Independent repetitions quantify operational variability and Monte Carlo error.
- Paired policy comparisons use the same random inputs and estimate staffing effects efficiently.
- Confidence intervals quantify finite-simulation error under a fixed model.
- More repetitions cannot repair unsupported input distributions or missing process mechanisms.

Next week: estimate and check the service-time distribution instead of assuming an exponential mean of six minutes.